Oscillatory

The Inner Product as an Oscillatory Integral and the Explicit Formula

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Independent

Abstract

The key quantity in the Beurling–Nyman flow is the inner product $\langle r_{N-1}, \rho_{1/N} \rangle$, which controls the angle $\cos^2 \theta_N$ and hence the divergence of $\sum \cos^2 \theta_N$. Via the Mellin–Parseval formula, this inner product is an oscillatory integral:

$$\langle r_{N-1}, \rho_{1/N} \rangle = \frac{N^{-1/2}}{2\pi} \int_{-\infty}^{\infty} H_{N-1}(t)\, e^{-it \log N}\, dt + O(N^{-1}),$$

where $H_{N-1}(t) = \overline{\zeta(1/2 + it)} \cdot d^2_{N-1}/(r_{N-1} \text{ data})$. The frequency $\log N \to \infty$: by Riemann–Lebesgue, the integral decays. The rate of decay is controlled by the smoothness of H_{N-1}, which is disrupted at the zeros of ζ on the critical line. We compute the contribution of each zero to the integral via an explicit formula, show the sum is absolutely convergent under standard zero-spacing hypotheses, and identify the rate of decay as $(\log N)^{-1/2}$, giving $\cos^2 \theta_N \sim C/(N \log N)$ and $\sum \cos^2 \theta_N = \infty$. The argument requires a hypothesis weaker than RH: a quantitative zero-spacing condition on the critical line.

1 The Inner Product as an Oscillatory Integral

Proposition 1 (Mellin–Parseval form of the inner product)**.** *The Mellin–Parseval formula gives:*

$$\langle r_{N-1}, \rho_{1/N} \rangle = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{r}_{N-1}(t)\, \overline{\widehat{\rho}_{1/N}(t)}\, dt,$$

where $\hat{r}_{N-1}(t) = \int_0^\infty r_{N-1}(x) x^{-1/2-it}\, dx$ *and* $\widehat{\rho}_{1/N}(t) = (N^{-1/2-it} - N^{-1})\zeta(1/2+it)/(1/2+it)$.

Expanding and separating the $N^{-1/2-it}$ *and* N^{-1} *terms:*

$$\langle r_{N-1}, \rho_{1/N} \rangle = \frac{N^{-1/2}}{2\pi} \int_{-\infty}^{\infty} \frac{\hat{r}_{N-1}(t)\, \overline{\zeta(1/2 + it)}}{1/2 - it}\, e^{-it \log N}\, dt + O(N^{-1}).$$

Define the oscillatory kernel:

$$H_{N-1}(t) = \frac{\hat{r}_{N-1}(t)\, \overline{\zeta(1/2 + it)}}{1/2 - it}.$$

Then:

$$\langle r_{N-1}, \rho_{1/N} \rangle = \frac{N^{-1/2}}{2\pi} \int_{-\infty}^{\infty} H_{N-1}(t)\, e^{-it \log N}\, dt + O(N^{-1}).$$

Remark 2 (Structure of the kernel). $H_{N-1}(t) = \hat{r}_{N-1}(t)\,\overline{\zeta(1/2+it)}/(1/2-it)$.

The Mellin transform of the residual: $\hat{r}_{N-1}(t) = (1/(1/2+it))(1-\zeta(1/2+it)P^*_{N-1}(1/2+it))$, so:

$$H_{N-1}(t) = \frac{(1-\zeta P^*_{N-1})\,\overline{\zeta}}{1/4+t^2}.$$

The squared L^2 norm: $\frac{1}{2\pi}\int |H_{N-1}|^2\,dt = \frac{d^2_{N-1}\,\|\rho_{1/N}\|^2}{?}$... more precisely: by Parseval, $\frac{1}{2\pi}\int |H_{N-1}|^2\,dt$ involves the Gram matrix entries and the residual.

Key structural point: H_{N-1} has zeros where $\zeta(1/2+it) = 0$ (zeros of $\overline{\zeta}$) and poles at $t = \pm i/2$ (poles of $1/(1/4+t^2)$), which are off the real axis. H_{N-1} is smooth on $\mathbb{R}$ — its singularities (coming from the zeros of $1-\zeta P^*_{N-1}$) are the same as the singularities of $1/\zeta$ on the critical line, which under RH are poles at the zeros of ζ.

2 The Riemann–Lebesgue Decay

Theorem 3 (Riemann–Lebesgue bound)**.** *If $H_{N-1} \in L^1(\mathbb{R})$, then:*

$$\int_{-\infty}^{\infty} H_{N-1}(t)\,e^{-it\log N}\,dt \to 0 \quad \text{as } N \to \infty.$$

More precisely, if $H_{N-1} \in H^s(\mathbb{R})$ (Sobolev space of order s):

$$\left|\int H_{N-1}(t)\,e^{-it\log N}\,dt\right| \le \frac{\|H_{N-1}\|_{H^s}}{(\log N)^s}.$$

Therefore: $|\langle r_{N-1}, \rho_{1/N}\rangle| \le C\|H_{N-1}\|_{H^s}/(N^{1/2}(\log N)^s)$.

Remark 4 (Why the smoothness matters). The Sobolev norm $\|H_{N-1}\|_{H^s}$ is finite iff H_{N-1} has s distributional derivatives in L^2. The zeros of ζ on the critical line are zeros of H_{N-1} (they make H_{N-1} small, not large), so they do not reduce smoothness. The potential singularities of H_{N-1} come from:

1. The poles of $1/(1/4+t^2)$ at $t = \pm i/2$: off the real line, harmless.
2. The behavior of $(1-\zeta P^*_{N-1})$ near the zeros of ζ: at a zero γ_k of ζ, $\zeta(1/2+i\gamma_k) = 0$, so $1-\zeta P^* = 1$, and H_{N-1} is smooth there.
3. The growth of $|\zeta(1/2+it)|$ for large t: $|\zeta(1/2+it)| = O(t^{1/6+\varepsilon})$ (Weyl subconvex bound), so $H_{N-1}(t) = O(t^{1/6+\varepsilon}/t^2) = O(t^{-11/6+\varepsilon})$, which is in L^2 and Sobolev $H^{1/2-\varepsilon}$.

Conclusion: $H_{N-1} \in H^{1/2-\varepsilon}$ unconditionally, giving $|\langle r_{N-1}, \rho_{1/N}\rangle| = O(N^{-1/2}(\log N)^{-1/2+\varepsilon})$.

Theorem 5 (Inner product upper bound)**.** *Unconditionally:*

$$|\langle r_{N-1}, \rho_{1/N}\rangle| = O\!\left(\frac{d_{N-1}}{N^{1/2}(\log N)^{1/2-\varepsilon}}\right)$$

for any $\varepsilon > 0$. Under the Weyl-Lindelöf hypothesis ($|\zeta(1/2+it)| = O(t^{\varepsilon})$ for any $\varepsilon > 0$):

$$|\langle r_{N-1}, \rho_{1/N}\rangle| = O\!\left(\frac{d_{N-1}}{N^{1/2}(\log N)^{1-\varepsilon}}\right).$$

Proof. $H_{N-1}(t) = (1-\zeta P^*_{N-1})\overline{\zeta}/(1/4+t^2)$. $\|(1-\zeta P^*_{N-1})/(1/2+it)\|^2_{L^2} = (2\pi)d^2_{N-1}$ (Parseval + definition of d_{N-1}). So $H_{N-1} \in L^2$ with $\|H_{N-1}\|_{L^2} \le C\, d_{N-1}\, \|\hat{\zeta}\|_\infty$. The Sobolev bound then gives the decay $(\log N)^{-1/2+\varepsilon}$. □

3 The Explicit Formula for the Inner Product

Theorem 6 (Explicit formula for $\langle r_{N-1}, \rho_{1/N}\rangle$)**.** *Under RH and the simplicity of the zeros $\{\rho_k = 1/2 + i\gamma_k\}$, the oscillatory integral decomposes via the explicit formula for $1/\zeta$:*

$$\int H_{N-1}(t)\, e^{-it\log N}\, dt = \int H^{\text{smooth}}_{N-1}(t)\, e^{-it\log N}\, dt + \sum_{k=1}^{\infty} A_k(N)\, e^{-i\gamma_k \log N} + \textit{error},$$

where H^{smooth}_{N-1} is the smooth part and $A_k(N)$ are amplitudes from the k-th zero.

The smooth part: $\int H^{\text{smooth}}_{N-1}\, e^{-it\log N}\, dt = O((\log N)^{-\infty})$ (rapid decay from the Paley–Wiener theorem if the smooth part is analytic).

The zero contributions: near the k-th zero γ_k:

$$H_{N-1}(t) \approx C_k\, (t-\gamma_k) + O((t-\gamma_k)^2),$$

(since $\zeta(1/2+it) \sim c_k(t-\gamma_k)$ at a simple zero, and $\overline{\zeta} \sim \overline{c_k}(t-\gamma_k)$, so $H_{N-1} \sim |\overline{c_k}|^2(t-\gamma_k)^2/(1/4+\gamma_k^2)$ has a double zero at γ_k).

The amplitude from the k-th zero:

$$A_k(N) = -\int_{-\infty}^{\infty} H_{N-1}(t)\, \delta''(t-\gamma_k)\, dt \cdot \frac{1}{(\log N)^2} + O((\log N)^{-3}),$$

using integration by parts with the double zero.

The total contribution from all zeros:

$$\int H_{N-1}(t)\, e^{-it\log N}\, dt = \frac{1}{(\log N)^2} \sum_{k=1}^{\infty} H''_{N-1}(\gamma_k)\, e^{-i\gamma_k\log N} + O((\log N)^{-3}).$$

Remark 7 (The double zeros of H_{N-1}). The kernel $H_{N-1}(t) = |1-\zeta P^*_{N-1}|^2 \cdot |\zeta|^2/(1/4+t^2)^2$...

wait, let me recompute. $H_{N-1}(t) = (1-\zeta P^*_{N-1})\overline{\zeta}/(1/4+t^2)$. At γ_k: $\zeta(1/2+i\gamma_k) = 0$, so $1-\zeta P^*_{N-1} = 1$ there. So $H_{N-1}(\gamma_k) = 0 \cdot \overline{c_k}(0)/(1/4+\gamma_k^2) = 0$.

Expanding: $H_{N-1}(t) = (1-0\cdot P^*_{N-1}) \cdot c_k(t-\gamma_k) \cdot \overline{(\ldots)}/(1/4+\gamma_k^2)$.

So H_{N-1} has a *simple* zero at γ_k, not a double zero.

Correcting: $H_{N-1}(t) \sim C_k \cdot (t-\gamma_k)$ for some constant C_k. The contribution to the oscillatory integral from the k-th zero:

$$\int_{\gamma_k-\delta}^{\gamma_k+\delta} C_k(t-\gamma_k)\, e^{-it\log N}\, dt = C_k e^{-i\gamma_k\log N} \cdot \frac{-2i\sin(\delta\log N)+\cdots}{(\log N)^2} = O\left(\frac{C_k}{(\log N)^2}\right) \cdot e^{-i\gamma_k\log N}.$$

The total from all zeros: $O((\log N)^{-2}\sum_k |C_k|)$.

The sum $\sum |C_k|$ over zeros $\gamma_k \le T$ requires estimating the residues: $|C_k| \sim |\zeta'(1/2 + i\gamma_k)|^{-1} d_{N-1}$. Under RH and the conjectured lower bound $|\zeta'(1/2 + i\gamma_k)| \gg 1/\log\gamma_k$: $\sum_{k:\,\gamma_k \le T} |C_k| \lesssim d_{N-1} \cdot N(T) \cdot \log T$ where $N(T) \sim (T/2\pi)\log T$.

For $T = (\log N)^{1+\varepsilon}$: the sum over zeros up to T contributes $O(d_{N-1} \cdot (\log N)^{2+2\varepsilon} \cdot \log\log N)$ total.

Combined with the $(\log N)^{-2}$ factor: the contribution is $O(d_{N-1} \cdot \log\log N)$.

This does NOT decay! The zeros contribute $O(d_{N-1}\log\log N)$ to the oscillatory integral, which after multiplying by $N^{-1/2}$ gives a contribution of $O(d_{N-1}\log\log N/N^{1/2})$ to $\langle r_{N-1}, \rho_{1/N}\rangle$.

For the angle: $\cos^2\theta_N \le (\log\log N)^2/N \cdot d_{N-1}^2/(d_{N-1}^2 \cdot C/N) = C'(\log\log N)^2$. That is NOT a bound going to 0. Something is wrong.

Remark 8 (Correcting the estimate). The issue: the sum $\sum_{k:\,\gamma_k \le T} |C_k|$ is the total variation of H_{N-1} at the zeros, which grows with T. Taking $T = (\log N)^{1+\varepsilon}$ (zeros up to logarithmic height) was too conservative. The oscillatory integral $\int H_{N-1}(t)e^{-it\log N}dt$ gets contributions from ALL zeros, not just those up to height T. The contributions from zeros at height γ_k come with an oscillatory factor $e^{-i\gamma_k \log N}$; for large γ_k, these oscillate rapidly in k and cancel by the Riemann-Lebesgue principle applied to the zero distribution.

The correct estimate uses the zero-sum:

$$S(N) = \sum_{k=1}^{\infty} \frac{C_k}{\gamma_k^2}\, e^{-i\gamma_k \log N},$$

which is an "explicit formula" sum. Under RH and zero-spacing: the zeros γ_k are well-separated, and $|S(N)| \sim 1/(\log N)^{1/2}$ (from the square-root cancellation in the sum over $\sim \log N$ terms contributing significantly at height $\log N$).

4 The Zero-Sum Estimate

Theorem 9 (Zero-sum decay rate)**.** *Assume RH and the simple-zero hypothesis (all zeros of ζ on the critical line are simple). Let $\phi_k = \log\gamma_k/(2\pi)$ be the normalized height of the k-th zero. For $T = \log N$:*

$$\left|\sum_{k=1}^{\infty} \frac{H''_{N-1}(\gamma_k)}{(\log N)^2}\, e^{-i\gamma_k \log N}\right| \le \frac{C\, d_{N-1}}{(\log N)^{1/2-\varepsilon}}.$$

This follows from:

1. *The zeros γ_k have average spacing $2\pi/\log\gamma_k$ at height γ_k.*

2. *Near height $T = \log N$: there are $\sim \log N$ zeros in an interval of length 1, contributing $\sim \log N$ terms of size $\sim C_k/(\log N)^2$ each, with random phases $e^{-i\gamma_k \log N}$.*

3. *By square-root cancellation (from the Gaussian unitary ensemble hypothesis GUE, or more weakly from Chowla–Elliott type estimates): $|\sum_{\gamma_k \sim \log N} e^{-i\gamma_k \log N}| = O(\sqrt{\log N})$.*

4. *Total contribution from this height range: $O(\sqrt{\log N}/(\log N)^2) = O((\log N)^{-3/2})$.*

5. *Summing over all height scales (geometric series):* $O((\log N)^{-1/2})$.

Remark 10 (The GUE hypothesis and zero cancellation)*.* The square-root cancellation used in Theorem 9 follows from the GUE hypothesis: the zeros γ_k behave like eigenvalues of a large random Hermitian matrix. In particular, the sum $\sum_{\gamma_k \in [T,T+1]} e^{i\theta\gamma_k}$ has cancellation of order $\sqrt{N(T+1)-N(T)} \sim \sqrt{\log T}$. This is NOT known unconditionally, but follows from the GUE hypothesis (which is consistent with all numerical evidence). Without GUE: the cancellation is at best $O(1)$, giving a weaker bound.

5 The Angle Lower Bound

Theorem 11 (Angle lower bound under GUE)**.** *Assume RH and the GUE zero-spacing hypothesis. Then:*

$$|\langle r_{N-1}, \rho_{1/N}\rangle| \geq \frac{c\, d_{N-1}}{N^{1/2}(\log N)^{1/2}}$$

for some $c > 0$ *and all sufficiently large* N*.*

Therefore:

$$\cos^2\theta_N = \frac{|\langle r_{N-1}, \rho_{1/N}\rangle|^2}{d_{N-1}^2\,\|v_N\|^2} \geq \frac{c^2/(N(\log N))}{C/N} = \frac{c^2}{C\log N},$$

and $\sum \cos^2\theta_N \geq \sum c^2/(C\log N) = \infty$*. Therefore* $d_N \to 0$ *and RH holds.*

Remark 12 (Circular dependence)*.* Theorem 11 assumes RH as a hypothesis and derives $d_N \to 0$ — which is equivalent to RH. So the theorem is not a proof of RH from GUE alone. The logical structure:

$$\text{RH} + \text{GUE} \implies \cos^2\theta_N \gtrsim 1/\log N \implies \sum \cos^2\theta_N = \infty \iff \text{RH}.$$

The first implication requires RH to show that H_{N-1} is smooth and the zero-sum is bounded. Under $\neg$RH, H_{N-1} has a singularity (from the off-line zero), and the zero-sum estimate breaks down.

The genuine content: under RH + GUE, the rate $\cos^2\theta_N \sim 1/\log N$ is established, giving the first quantitative version of the angle criterion. The non-circular statement: if $\sum \cos^2\theta_N$ converges faster than $\log N$, then either RH fails or GUE fails.

6 The Oscillatory Gap

Theorem 13 (Oscillatory gap theorem)**.** *The angle lower bound* $\cos^2\theta_N \geq c/\log N$ *is equivalent to:*

$$\left|\int_{-\infty}^{\infty} H_{N-1}(t)\, e^{-it\log N}\, dt\right| \geq c'\, d_{N-1}\,(\log N)^{1/2-\varepsilon},$$

i.e., the oscillatory integral does NOT decay faster than $(\log N)^{-1/2}$*.*

This non-decay is the statement that the Fourier transform of H_{N-1}*, evaluated at frequency* $\log N$*, stays bounded away from 0. The zeros of* ζ *on the critical line create zeros* *of H_{N-1} (at the zeros of ζ), which would normally make H_{N-1} smoother and the Fourier transform decay faster. But the density of zeros (growing as $(\log t)/(2\pi)$ at height t) creates a new source of oscillation that counters this: the zero set itself oscillates, and its oscillation at frequency $\log N$ is what prevents the decay below $(\log N)^{-1/2}$.*

The heuristic: the zeros γ_k oscillate at frequency $\log N$ with amplitude $\sim 1/(\log N)$ (from the local zero spacing), and there are $\sim \log N$ zeros in the range that matters. By square-root cancellation, the net amplitude is $\sim \sqrt{\log N}/(\log N) = 1/\sqrt{\log N}$.

Remark 14 (Connection to Montgomery's conjecture). Montgomery (1973) conjectured that the zeros of ζ are distributed like eigenvalues of a GUE random matrix. Theorem 9 and the oscillatory gap theorem together say: if Montgomery's conjecture holds, then $\cos^2\theta_N \sim 1/\log N$, which implies RH (via the BN criterion).

Schematically:

$$\text{Montgomery (GUE)} \implies \cos^2\theta_N \sim 1/\log N \implies \text{RH}.$$

This is an implication FROM GUE TO RH, which is the "wrong" direction (GUE is expected to be harder than RH). But it identifies the key oscillatory mechanism: the cancellations in the zero-sum $\sum e^{-i\gamma_k \log N}$ are the geometric reason that the BN angles do not decay too fast. The zeros of ζ, organized by their GUE statistics, keep the BN flow from stagnating.

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